# Expanding Knowledge Boundaries via LLM-Grounded Alignment for Drug Combination Recommendation

Tengfei Ma*
College of Computer Science and Electronic Engineering, Hunan University
Changsha, Hunan, China
tfma@hnu.edu.cn

Yuqin He*
College of Computer Science and Electronic Engineering, Hunan University
Changsha, Hunan, China
yuqinhe@hnu.edu.cn

Zhonghao Ren
College of Computer Science and Electronic Engineering, Hunan University
Changsha, Hunan, China
zhonghaoren98@gmail.com

Bosheng Song†
College of Computer Science and Electronic Engineering, Hunan University
Changsha, Hunan, China
Yuelushan Laboratory
Changsha, Hunan, China
boshengsong@hnu.edu.cn

Qian Li
School of Computer Science, Beijing University of Posts and Telecommunications
Beijing, China
li.qian@bupt.edu.cn

Xiangxiang Zeng†
State Key Laboratory of Chemo and Biosensing, College of Computer Science and Electronic Engineering, Hunan University
Changsha, Hunan, China
Yuelushan Laboratory
Changsha, Hunan, China
xzeng@hnu.edu.cn

## Abstract

Combination therapy is widely used to treat complex diseases, yet predicting synergistic drug combinations remains challenging for rare and underrepresented cell lines due to severe data sparsity. Existing methods rely primarily on observed drug–cell responses and learn representations within a limited experimental knowledge regime, resulting in poor generalization in cold-start and long-tail settings. To address this limitation, we propose L$^2$aCo, a model-agnostic knowledge alignment framework that leverages large language models as external biomedical knowledge sources to expand the effective knowledge boundary for drug combinations. L$^2$aCo augments drug and cell-line representations with LLM-inferred semantic profiles capturing biological mechanisms and contextual relations beyond experimental measurements. These semantic representations are integrated with conventional molecular and cellular features through a representation-level alignment mechanism, which injects relational knowledge while preserving task-relevant experimental signals. By enabling effective knowledge transfer under data scarcity, L$^2$aCo substantially improves generalization for cold-start and long-tail cell lines. Experiments on multiple public benchmarks show consistent and significant performance gains for novel cell lines when L$^2$aCo is integrated with drug synergy predictors, with pronounced improvements on novel cell lines. Code is available at: https://github.com/xiaomingaaa/LaCo.

*Both authors contributed equally to this work.
†Corresponding Authors


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## CCS Concepts

• **Applied computing** → Bioinformatics.

## Keywords

Drug Combination Prediction, Large Language Model

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## 1 Introduction

Combination therapy, which involves the simultaneous administration of multiple therapeutic agents, has emerged as a powerful strategy for treating complex diseases [25, 34]. Compared with monotherapy, drug combinations can enhance therapeutic efficacy while reducing adverse effects by enabling lower dosages of individual drugs [2, 16, 33]. As a result, identifying effective and synergistic drug combinations has become a critical problem in modern precision medicine. However, despite extensive research efforts, accurately predicting synergistic drug combinations remains a challenging task.

Early attempts to discover novel drug combinations primarily relied on clinical trials, which are time-consuming, costly, and may expose patients to ineffective or even harmful treatments [32]. With the increasing availability of drug combination datasets [24, 40], recent research has shifted toward developing computational approaches to efficiently predict unknown drug combinations [5, 14]. These methods aim to prioritize promising combinations before clinical validation, thereby reducing experimental costs and risks. A large body of existing computational works focuses on learning predictive models from experimentally derived features of drugs

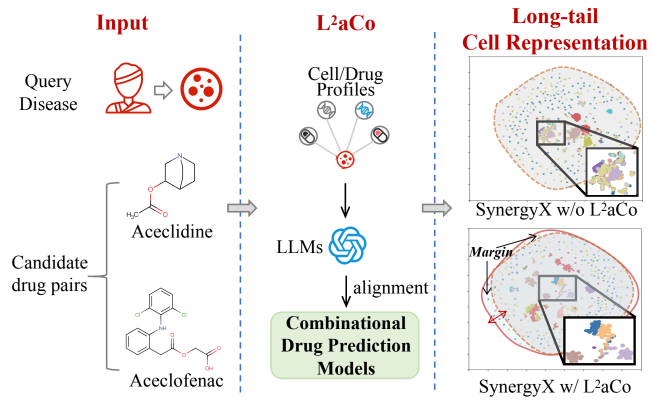


**Figure 1: Existing methods fail to properly cluster rare cell lines, while L$^2$aCo leverages LLM-derived knowledge to expand the representation boundary and improve coverage of long-tail cells.**

and cells [19, 30]. Typical approaches represent drugs using handcrafted physicochemical descriptors or molecular fingerprints, and characterize cell lines using gene expression profiles. These representations are often concatenated and fed into deep neural networks for synergy prediction. To improve representation quality, recent studies explore more expressive modeling techniques, such as graph neural networks (GNNs) for capturing molecular topological structures [21, 36, 37], multi-channel architectures to enhance drug–drug interactions [3, 9, 43], and drug-induced gene expression profiles to strengthen drug–cell associations [1, 15, 37].

Despite their promising performance, these approaches are fundamentally constrained by the scope of available experimental data. In particular, they primarily model interactions among observed biomedical entities (e.g., drugs, genes, and cell lines) and struggle to generalize in data-scarce scenarios, such as cold-start or long-tail cell lines (Figure 1). More importantly, they lack explicit mechanisms to incorporate broader biomedical knowledge that governs drug interactions and cellular responses, which is crucial for understanding and predicting combinatorial effects. To address these limitations, recent studies introduce biomedical knowledge graphs (KGs) to augment the representations of drugs and cells [31, 41]. By encoding structured relationships among biomedical entities, KG-based methods provide a principled way to inject external knowledge into predictive models. However, real-world biomedical KGs are often incomplete, noisy, and expensive to curate and maintain. Their limited coverage and static nature restrict their effectiveness, especially when dealing with rare drugs, novel cell lines, or emerging biomedical knowledge.

In this work, we propose L$^2$aCo, a model-agnostic knowledge alignment framework for drug combination prediction that unifies structured experimental features and unstructured biomedical knowledge within a shared representation space. Rather than relying on LLM-derived text alone, L$^2$aCo emphasizes structured and unstructured interaction modeling, enabling complementary information sources to jointly expand the effective knowledge boundary under data scarcity. L$^2$aCo utilizes LLMs to infer auxiliary semantic profiles for drugs and cell lines from unstructured biomedical text, capturing functional mechanisms, biological contexts, and relational cues that are often missing in structured resources. To integrate such unstructured knowledge with experimental signals, L$^2$aCo introduces a representation-level alignment mechanism that aligns LLM-derived semantics with conventional molecular, cellular, and KG-based features and injects relational information into downstream predictors. By expanding the effective knowledge boundary of representation learning, L$^2$aCo enables drug combination models to capture context-aware combinatorial preferences conditioned on cellular states, while preserving task-relevant experimental specificity. The contributions of L$^2$aCo include:

- We show that poor generalization on cold-start and long-tail cell lines arises from learning within a restricted experimental knowledge boundary, and improve drug combination prediction under data scarcity via knowledge boundary expansion.
- We propose L$^2$aCo, a model-agnostic LLM-grounded knowledge alignment framework that enriches drug and cell representations with LLM-inferred biomedical semantics via representation-level alignment, and can be seamlessly integrated into existing drug combination predictors.
- We provide theoretical analysis on hypothesis space expansion and extensive experiments demonstrating consistent performance gains, robustness to knowledge perturbations, and substantial improvements in cold-start and long-tail settings.

# 2 Related Work

## 2.1 Drug Combination Prediction

Drug combination and synergy prediction have been extensively studied as a core problem in precision medicine. With the increasing availability of high-throughput drug combination datasets [13, 24], computational methods for drug synergy prediction have attracted growing attention [9, 26]. Early deep learning–based approaches typically adopt a two-branch architecture, where drug features and cell line features are extracted separately and then concatenated for prediction [31]. Representative models such as DeepSynergy [30] and MatchMaker [19] utilize physicochemical descriptors for drugs and gene expression profiles for cell lines, followed by fully connected networks to model synergy outcomes. Subsequent studies improve upon this paradigm by incorporating more expressive feature extractors [15, 38], including graph neural networks (GNNs) for molecular structure modeling [10, 45] and drug-induced gene expression profiles to enhance drug–cell associations [17, 18, 35]. While these approaches demonstrate strong predictive performance, they largely treat drug–drug and drug–cell interactions implicitly through feature concatenation. As a result, they may fail to capture fine-grained interaction patterns that are critical for understanding combinatorial effects, particularly under complex biological contexts [6, 20, 41]. In contrast, our work leverages LLMs as a flexible and extensible knowledge source and introduces a representation-level alignment framework that augments existing drug combination predictors without architectural modification, enabling improved generalization under data scarcity.

## 2.2 Knowledge-Enhanced Learning

To better capture biological interactions among drugs and cell lines, a growing body of work has explored knowledge-enhanced learning approaches that incorporate external biomedical information into representation learning [6, 39]. These methods typically integrate structured knowledge sources, such as ontologies [27] and biomedical knowledge graphs [22], through multi-view or cross-view learning frameworks, leading to more informative and robust representations [41]. Despite their effectiveness, such approaches heavily rely on curated and structured knowledge, which is often incomplete, domain-specific, and costly to construct and maintain [7]. Recently, large language models (LLMs) have demonstrated strong capabilities in capturing and synthesizing biomedical and chemical knowledge from large-scale unstructured text corpora [4, 8, 28]. As a result, LLMs have been increasingly applied to scientific tasks such as molecular understanding [8], drug discovery [44], and biomedical question answering [42]. In the context of drug synergy prediction, SynerGPT introduces a novel formulation of in-context drug synergy learning, in which GPT-style models are pretrained to directly model drug synergy functions from textual and structural inputs [7]. While promising, these approaches primarily treat LLMs as standalone predictors or generators, and their integration with existing drug synergy models remains limited.

In contrast, our work adopts a different perspective by leveraging LLMs as an *model-agnostic knowledge alignment framework*. We use LLMs to infer auxiliary semantic knowledge for drugs and cell lines from unstructured text and integrate this knowledge into existing drug combination predictors through a unified representation augmentation framework. This design enables effective knowledge transfer from unstructured textual knowledge to structured prediction models and significantly improves generalization, particularly in cold-start and long-tail scenarios.

# 3 Method

## 3.1 Preliminaries

**Problem Definition.** We study the problem of drug combination recommendation. Let $d_i \in \mathcal{D}$ denote a drug, $c \in C$ denote a cell line, and $y_{ijc} \in \mathbb{R}$ denote the observed synergy score of drug pair $(d_i, d_j)$ on cell line $c$. Each drug $d$ is associated with a raw feature representation $\mathbf{x}_d \in \mathbb{R}^p$, such as molecular fingerprints or graph-based encodings. Each cell line $c$ is represented by experimental measurements $\mathbf{x}_c \in \mathbb{R}^q$, e.g., gene expression profiles. Existing drug combination models learn a predictor

$$f_\theta(\mathbf{x}_{d_i}, \mathbf{x}_{d_j}, \mathbf{x}_c) \rightarrow \hat{y}_{ijc}, \tag{1}$$

where $f_\theta$ can be instantiated as a feature-based model or a graph neural network. In cold-start or long-tail settings, the experimental representation $\mathbf{x}_c$ is often sparse or insufficient, which significantly limits generalization. Our goal is to enhance representation learning by incorporating external knowledge beyond experimental data.

**Rare and Long-Tail Cell Lines.** In real-world datasets, cell lines typically follow a long-tail distribution, where a small number of well-studied cell lines dominate the training data, while many cell lines have only limited observations. We refer to cell lines with insufficient training samples as *rare* or *long-tail* cell lines. For such cell lines, experimental representations $\mathbf{x}_c$ are often sparse or noisy, which leads to high representation bias and poor generalization when learning from experimental data alone.

## 3.2 Overview of $\mathrm{L}^2$aCo

The proposed framework, termed **$\mathbf{L}^2$aCo**, augments existing drug combination prediction models by injecting LLM-derived semantic knowledge into representation learning. As illustrated in Figure 1, $\mathrm{L}^2$aCo operates as a model-agnostic plug-in that enhances drug and cell representations without modifying downstream prediction architectures. Specifically, $\mathrm{L}^2$aCo first constructs knowledge-aware queries for drugs and cells lines to elicit semantic information from a pretrained LLM. The resulting semantic representations are then aligned with experimental representations through cross-view learning. Finally, the enhanced representations are fed into the existing drug combination predictors, such as feature-based models or graph neural networks.

## 3.3 Representation of Biological Knowledge

**Reliable Knowledge Query Construction.** To extract semantic knowledge relevant to drug combination prediction, $\mathrm{L}^2$aCo constructs structured and interactive queries for drugs and cell lines.

For a drug $d$, the query $Q_d$ includes its name, known indications, molecular mechanisms, and interaction-related facts to enhance the reliability of the generated text. For a cell line $c$, the query $Q_c$ includes cell identity, gene expression characteristics, and over- or under-expressed biomarkers. Formally, we define

$$Q_d = \mathrm{Prompt}(d, \mathcal{K}_d), \quad Q_c = \mathrm{Prompt}(c, \mathcal{K}_c), \tag{2}$$

where $\mathcal{K}_d$ and $\mathcal{K}_c$ denote auxiliary biomedical knowledge sources. These queries are designed to encourage the LLM to generate concise semantic summaries and rationales that reflect the biological roles and functional contexts of drugs and cell lines. As an illustrative example, for the drug *Selumetinib*, the constructed query $Q_d$ contains its name, therapeutic indications, and known molecular mechanisms. Given this query, the LLM generates a concise semantic summary describing that Selumetinib inhibits the MAPK/ERK signaling pathway, along with a rationale highlighting its functional role in regulating cell proliferation. Similarly, for a cell line characterized by overexpression of specific genes, the query $Q_c$ prompts the LLM to summarize its transcriptional profile and biological context. These LLM-generated outputs are subsequently encoded as semantic representations, which are used for downstream alignment and prediction. We show more cases and prompts in Appendix A.

**LLM-based Semantic Representation Generation.** Given the constructed queries, we employ a frozen pretrained LLM to generate semantic summaries and rationales. The generated text is then encoded into dense semantic embeddings using the LLM's representation layer:

$$\mathbf{s}_d = \mathcal{M}_{\mathrm{LLM}}(Q_d), \quad \mathbf{s}_c = \mathcal{M}_{\mathrm{LLM}}(Q_c). \tag{3}$$

The semantic representations capture high-level biomedical knowledge that is not explicitly present in experimental measurements, such as functional relationships, biological processes, and mechanistic descriptions. Since the LLM is pretrained on large-scale biomedical corpora, these embeddings provide a rich and extensible source of external knowledge.

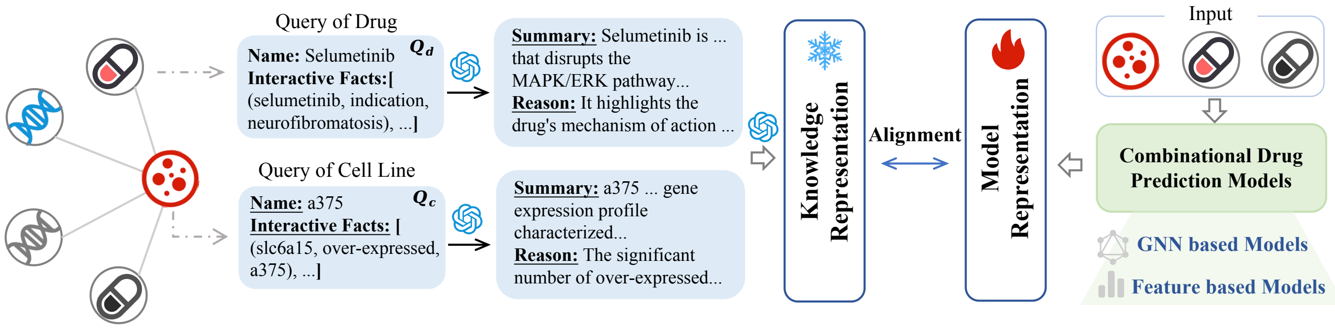


**Figure 2: L$^2$aCo leverages LLMs to infer semantic knowledge for drugs and cell lines and aligns it with model representations via representation-level alignment. This design expands the effective knowledge boundary of drug combination prediction models and improves generalization, particularly for cold-start and long-tail cell lines.**

## 3.4 Model-agnostic Knowledge Alignment

To integrate semantic knowledge with experimental representations, L$^2$aCo performs cross-view representation fusion. Let $\mathbf{z}_x$ and $\mathbf{z}_s$ denote the latent experimental and semantic embeddings obtained from their respective encoders. The fusion process consists of two steps: (i) aligning the two views through contrastive learning and KL regularization, and (ii) combining them into enhanced representations:

$$\hat{\mathbf{z}} = \mathrm{Fuse}(\mathbf{z}_x, \mathbf{z}_s), \tag{4}$$

where Fuse$(\cdot)$ can be implemented as concatenation, gated fusion, or attention-based aggregation. The resulting representation $\hat{\mathbf{z}}$ preserves experimental specificity while incorporating semantically grounded knowledge inferred by the LLM.

**Training Objective** The overall training objective of L$^2$aCo combines the task-specific prediction loss with cross-view alignment losses:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha\mathcal{L}_{\text{con}} + \beta\mathcal{L}_{\text{KL}}, \tag{5}$$

where $\mathcal{L}_{\text{task}}$ denotes the prediction loss for drug synergy, and $\mathcal{L}_{\text{con}}$ and $\mathcal{L}_{\text{KL}}$ correspond to the contrastive and KL-regularized objectives. The hyperparameters $\alpha$ and $\beta$ control the strength of semantic knowledge transfer. This joint objective ensures that semantic information is incorporated in a principled manner without overwhelming experimental signals.

**Integration with Existing Prediction Models** Given enhanced representations $\hat{\mathbf{x}}_d$ and $\hat{\mathbf{x}}_c$, L$^2$aCo augments existing predictors as

$$\hat{y}_{ijc} = f_\theta(\hat{\mathbf{x}}_{d_i}, \hat{\mathbf{x}}_{d_j}, \hat{\mathbf{x}}_c). \tag{6}$$

Since L$^2$aCo operates at the representation level, it can be seamlessly integrated with feature-based models, graph neural networks, and other combinational drug prediction architectures.

## 3.5 Analysis of Knowledge Expansion

We provide a formal analysis to explain why L$^2$aCo can significantly improve drug combination prediction, especially for rare or underrepresented cell lines, from a knowledge boundary perspective.

**Knowledge Boundary of Experimental Learning.** Let $\mathcal{X} = \{\mathbf{x}_d, \mathbf{x}_c\}$ denote the space of experimentally observed representations of drugs and cell lines, and let $\mathcal{H}_{\text{exp}}$ be the hypothesis class induced by predictors that operate solely on $\mathcal{X}$. We define the corresponding *experimental knowledge boundary* as

$$\mathcal{B}_{\text{exp}} = \mathrm{span}(\mathcal{X}), \tag{7}$$

which characterizes the maximal information accessible to the model from experimental data. For rare cell lines, the effective dimension of $\mathcal{B}_{\text{exp}}$ is severely limited due to data sparsity, leading to high approximation bias.

**Semantic Knowledge Boundary Induced by LLMs.** Let $\mathcal{S} = \{\mathbf{s}_d, \mathbf{s}_c\}$ denote semantic representations generated by a pretrained large language model. Since the LLM is trained on large-scale biomedical corpora independently of the experimental dataset, $\mathcal{S}$ encodes external relational and contextual knowledge. We define the *semantic knowledge boundary* as

$$\mathcal{B}_{\text{LLM}} = \mathrm{span}(\mathcal{S}). \tag{8}$$

In practice, $\mathcal{B}_{\text{LLM}}$ is not contained in $\mathcal{B}_{\text{exp}}$, as it captures biomedical semantics that are not recoverable from experimental measurements or curated knowledge graphs.

**Augmented Knowledge Boundary.** L$^2$aCo jointly leverages experimental and semantic representations through cross-view alignment. The resulting accessible knowledge boundary is given by

$$\mathcal{B}_{\text{aug}} = \mathrm{span}\left(\mathcal{B}_{\text{exp}} \cup \mathcal{B}_{\text{LLM}}\right), \tag{9}$$

which strictly contains $\mathcal{B}_{\text{exp}}$.

**Proposition 1 (Strict Boundary Expansion).** *If $\mathcal{B}_{\text{LLM}} \not\subseteq \mathcal{B}_{\text{exp}}$, then $\mathcal{B}_{\text{aug}} \supsetneq \mathcal{B}_{\text{exp}}$, and the hypothesis class induced by L$^2$aCo, denoted $\mathcal{H}_{\text{aug}}$, strictly contains $\mathcal{H}_{\text{exp}}$.*

**Corollary 1 (Bias Reduction for Rare Cell Lines).** *For rare or underrepresented cell lines, where* $\dim(\mathcal{B}_{\text{exp}})$ *is small, the relative expansion $\mathcal{B}_{\text{aug}}/\mathcal{B}_{\text{exp}}$ is larger, leading to a greater reduction in representation bias and improved generalization.*

This analysis shows that L$^2$aCo improves drug combination prediction not by merely fitting stronger models within a fixed information regime, but by expanding the effective knowledge boundary in a principled and regularized manner. This boundary expansion is particularly beneficial in data-scarce settings, which explains the empirical gains observed for long-tail cell lines.

## 4 Experiments

In this section, we evaluate the proposed L$^2$aCo[1] by considering the following *key research questions*:

- **Q1:** Can L$^2$aCo consistently improve different base models on novel cell lines?
- **Q2:** Does L$^2$aCo achieve larger gains on long-tail cell lines with limited training data?
- **Q3:** How do different LLMs and semantic perturbations affect the performance of L$^2$aCo?
- **Q4:** How does L$^2$aCo reshape learned representations and interact with noisy structured knowledge?

### 4.1 Experimental Settings

***Dataset.*** To comprehensively evaluate the effectiveness of our method, we conduct experiments on two widely used public benchmarks for drug synergy prediction. (i) **DrugComb** [40]: From the DrugComb database, we use 74,974 drug combination samples involving 736 drugs and 102 cell lines, including 40,158 synergistic and 34,816 antagonistic combinations. (ii) **DrugCombDB** [24]: DrugCombDB is a publicly released dataset containing 32,813 drug combination samples across 570 drugs and 46 cell lines, with 18,007 synergistic and 14,806 antagonistic combinations. Each sample in both datasets consists of two drugs, one cell line, and an associated synergy score. To rigorously evaluate generalization, we adopt a *cell line–level inductive split*, where cell lines in the validation and test sets are entirely disjoint from those in the training set. This protocol evaluates the model's transferability and robustness to unseen and underrepresented cell lines.

***Evaluation.*** We evaluate our method on two public drug combination benchmarks using three standard binary classification metrics: ACC, ROC-AUC, and ROC-PR. For each dataset, we adopt a *cell line–level inductive split*, where the cell lines in the validation and test sets are entirely disjoint from those in the training set. All results are reported as the average performance over five independent runs. ACC measures overall prediction correctness, ROC-AUC evaluates the model's ability to distinguish synergistic from antagonistic drug pairs, and ROC-PR emphasizes performance under class imbalance by characterizing the precision–recall trade-off.

***Implementation Details.*** We employ the GPT-5 model provided by OpenAI to automatically generate structured textual descriptions for drugs and cell lines, covering their mechanisms of action, therapeutic indications, and biological properties. The generated texts are then encoded using text-embedding-3-large [29] to obtain high-quality semantic embeddings for downstream modeling. All models are trained using the Adam optimizer and binary cross-entropy loss. We use a fixed batch size for all experiments and apply early stopping based on validation performance (e.g., ACC) to mitigate overfitting. All reported results are averaged over five independent runs with different random seeds.

***Baselines.*** We integrate the proposed method (L$^2$aCo) into multiple state-of-the-art drug synergy prediction models to assess its effectiveness and generalizability.

- **SynergyX** [9]: A graph-based framework that captures chemical and biological features of drug–cell pairs using multi-level attention and representation fusion.
- **DeepDDS** [37]: A deep learning model that combines drug graph convolutional networks and cell-line gene expression profiles, leveraging drug–target interaction signals for synergy prediction.
- **DFFNDDS** [38]: A dual-fusion framework that employs multi-view feature extraction and feature disentanglement to improve the generalizability and interpretability of synergy predictions.
- **HanSynergy** [6]: A heterogeneous graph attention network that explicitly models drug–cell–target–pathway relationships to capture complex biological dependencies across modalities.
- **KGNN** [23]: A knowledge graph neural network that incorporates structured biomedical knowledge (e.g., drug–target or drug–disease relations) to enhance drug synergy prediction.
- **LightGCN** [11]: A lightweight graph collaborative filtering model based on linear neighborhood aggregation on the interaction graph, providing a strong parameter-efficient baseline.
- **H-GAT** [12]: An attention-based multi-relational graph encoder that assigns adaptive importance to distinct node and edge types, enabling fine-grained modeling of cross-omic and pharmacological signals in drug–cell graphs.

### 4.2 Performance Comparison (Q1 & Q2)

*4.2.1* ***Performance under Novel Cell Lines (Q1).*** To evaluate the effectiveness of **L$^2$aCo** in improving drug synergy prediction performance on novel cell lines, we integrate it into several state-of-the-art synergy prediction models. Each experiment is run with five random seeds, and we report the average performance in Table 1. The results lead to several key observations:

(i) We consistently observe that integrating L$^2$aCo with different backbone models leads to improved performance compared to their original counterparts across both DrugComb and DrugCombDB. This trend holds for both feature-based backbones and graph-based backbones, suggesting that L$^2$aCo is a model-agnostic enhancement applicable across diverse synergy predictors. (ii) We attribute the observed improvements to two main factors. First, L$^2$aCo leverages large language models to encode semantic descriptions of drugs and cell lines, enriching model representations with complementary biological and pharmacological knowledge beyond conventional molecular and omics features. Second, L$^2$aCo facilitates effective alignment between LLM-derived semantic embeddings and backbone representations, enabling more accurate modeling of drug–cell interactions while mitigating irrelevant or noisy information.

### 4.3 Performance for Long-tail Cell Lines (Q2)

We further evaluate the generalization ability of L$^2$aCo on long-tail cell lines. As shown in Figure 3, the cell line frequency distributions in DrugComb and DrugCombDB exhibit pronounced long-tail characteristics, where a small number of cell lines account for a large proportion of samples, while the majority appear only infrequently.

[1]Code is available at: https://github.com/xiaomingaaa/LaCo
[2]Due to GPU memory and runtime constraints, we report the HANSynergy results averaged over two runs with different random seeds instead of five.

**Table 1: Performance comparison between original and augmented models on DrugComb and DrugCombDB datasets. Results of augmented models are reported as mean ± standard deviation over five random seeds.**

| Model | Setting | DrugComb | | | DrugCombDB | | |
|---|---|---|---|---|---|---|---|
| | | ACC | ROC-AUC | ROC-PR | ACC | ROC-AUC | ROC-PR |
| SynergyX | base | 60.66 ± 0.73 | 64.74 ± 0.62 | 66.10± 0.59 | 71.26± 0.67 | 77.99± 0.48 | 82.90± 0.73 |
| | $L^2$aCo | 62.06 ± 0.64 | 66.57 ± 0.57 | 67.64 ± 0.70 | 72.11 ± 0.50 | 78.01 ± 0.42 | 83.14 ± 0.67 |
| | **impr. (%)** | *+2.31%* | *+2.83%* | *+2.33%* | *+1.19%* | *+0.03%* | *+0.29%* |
| DFFNDDS | base | 58.93 ± 0.18 | 61.50 ± 0.09 | 63.13 ± 0.14 | 71.06 ± 0.23 | 76.56 ± 0.15 | 78.91 ± 0.54 |
| | $L^2$aCo | 59.86 ± 0.32 | 62.91 ± 0.36 | 63.85 ± 0.89 | 71.63 ± 0.39 | 77.45 ± 0.70 | 80.75 ± 1.23 |
| | **impr. (%)** | +1.58% | +2.29% | +1.14% | +0.80% | +1.16% | +2.33% |
| DeepDDS | base | 57.46 ± 2.76 | 62.92± 1.26 | 66.95 ± 1.85 | 65.65 ± 2.59 | 75.00 ± 2.42 | 66.71 ± 3.38 |
| | $L^2$aCo | 58.30 ± 2.61 | 64.36 ± 3.33 | 68.72 ± 0.48 | 71.95 ± 0.41 | 78.22 ± 3.17 | 70.13 ± 3.11 |
| | **impr. (%)** | +1.46% | +1.44% | +1.77% | +9.60% | +4.29% | +5.13% |
| HANSynergy[2] | base | 60.61 ± 1.70 | 64.37 ± 1.39 | 64.40 ± 0.61 | 71.17 ± 0.50 | 78.26 ± 0.52 | 82.28 ± 0.11 |
| | $L^2$aCo | 61.22 ± 1.01 | 65.46 ± 0.93 | 64.89 ± 0.12 | 72.38 ± 0.56 | 79.17 ± 0.64 | 83.32 ± 0.73 |
| | **impr. (%)** | +0.19% | +1.18% | +0.23% | +1.70% | +0.16% | +1.26% |
| KGNN | basse | 59.89 ± 0.09 | 62.88 ± 0.08 | 63.23 ± 0.12 | 65.08 ± 0.57 | 71.90 ± 0.48 | 76.70 ± 0.50 |
| | $L^2$aCo | 60.56 ± 0.46 | 65.05 ± 0.41 | 66.78 ± 0.43 | 69.33 ± 0.82 | 76.53 ± 0.85 | 80.76 ± 0.94 |
| | **impr. (%)** | +1.12% | +3.45% | +5.61% | +6.53% | +6.44% | +5.29% |
| LightGCN | base | 59.88 ± 0.89 | 62.88 ± 0.08 | 63.23 ± 0.12 | 65.67 ± 0.09 | 71.28 ± 0.07 | 76.18 ± 0.10 |
| | $L^2$aCo | 60.13 ± 0.34 | 63.83 ± 0.59 | 65.47 ± 0.95 | 70.30 ± 0.83 | 76.71 ± 1.07 | 80.53 ± 1.07 |
| | **impr. (%)** | +0.42% | +1.51% | +3.54% | +7.05% | +7.62% | +5.71% |
| H-GAT | base | 58.22 ± 0.53 | 62.56 ± 0.15 | 62.73 ± 0.21 | 66.04 ± 0.41 | 72.31 ± 0.49 | 77.27 ± 0.27 |
| | $L^2$aCo | 60.35 ± 0.52 | 64.99 ± 0.57 | 66.51 ± 0.72 | 67.84 ± 0.47 | 73.98 ± 0.77 | 78.18±0.51 |
| | **impr. (%)** | +3.66% | +3.88% | +6.03% | +2.73% | +2.31% | +1.18% |

Notably, the two datasets present different degrees of data imbalance, highlighting the need for dataset-specific evaluation settings. To account for these distributional differences, we adopt dataset-specific long-tail evaluation splits. For each dataset, cell lines with the lowest occurrence frequencies are treated as low-resource targets, and samples associated with these cell lines are reserved for testing, while the remaining samples are used for training. This setting reflects realistic scenarios in which models are required to generalize to cell lines with scarce or limited historical observations. We select four representative base models (SynergyX, DFFNDDS, KGNN, and LightGCN) and evaluate $L^2$aCo under the long-tail cell-line setting. As shown in Figure 4, on both DrugComb and DrugCombDB, $L^2$aCo consistently outperforms the corresponding base models and yields gains on both ROC-AUC and ROC-PR, indicating stronger robustness and generalization under data-scarce and imbalanced conditions. We attribute the improvements to two factors: (i) $L^2$aCo augments base models with LLM-encoded semantic representations of cell lines and drugs, reducing dependence on extensive cell-line-specific supervision and improving performance on unseen or low-frequency cell lines; and (ii) it promotes alignment between semantic embeddings and base representations, facilitating effective fusion of semantic cues and mitigating irrelevant noise, as reflected by consistent gains in ROC-PR.

## 4.4 Robustness of $L^2$aCo (Q3 & Q4)

*4.4.1* ***Sensitivity to Different LLMs (Q3).*** To assess the sensitivity of $L^2$aCo to the choice of profiling LLM, we replace the LLM used to generate drug and cell-line profiles while keeping the prompts, embedding encoder, and all training and evaluation protocols unchanged (Table 2). Across KGNN, LightGCN, and H-GAT, $L^2$aCo consistently outperforms the corresponding base models when using GPT-3.5, GPT-4o-mini, or GPT-5, indicating that its effectiveness does not rely on a specific LLM, but rather on the general design of semantic injection and cross-view alignment. Meanwhile, profiling quality affects the magnitude of the gains: stronger LLMs typically yield larger improvements, particularly on ranking-oriented metrics such as ROC-AUC and ROC-PR. This suggests that higher-quality profiles provide more informative semantic signals for synergy ranking, while the alignment mechanism helps preserve stable benefits even when profiling is less accurate.

*4.4.2* ***Performance of LLM embeddings (Q3).*** Table 3 compares standalone LLM embeddings with $L^2$aCo under different augmentation settings. Overall, using LLM embeddings alone yields reasonable performance, indicating that LLMs capture useful semantic information about drugs and cell lines. For example, GPT-5 achieves an AUC/AUPR of 61.24/63.07 on DrugComb and 73.68/78.68 on DrugCombDB. However, directly relying on LLM embeddings is insufficient to model complex drug–drug–cell interactions. When

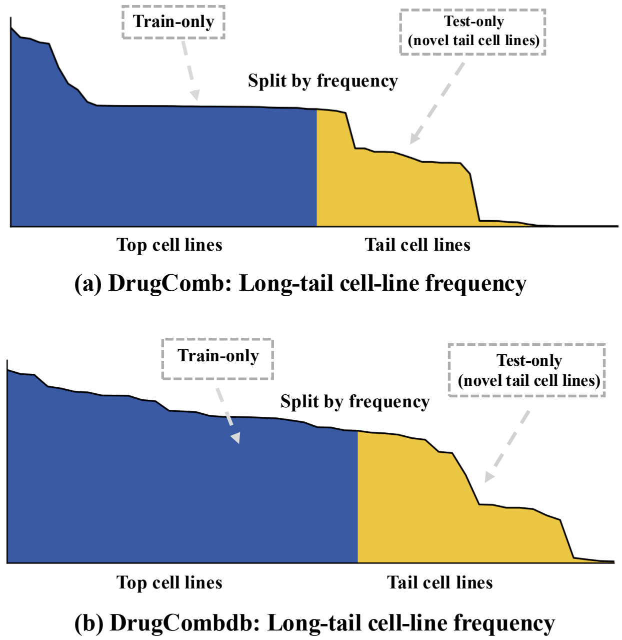


**Figure 3: Dataset-specific novel cell-line splits based on long-tail frequency distributions.**

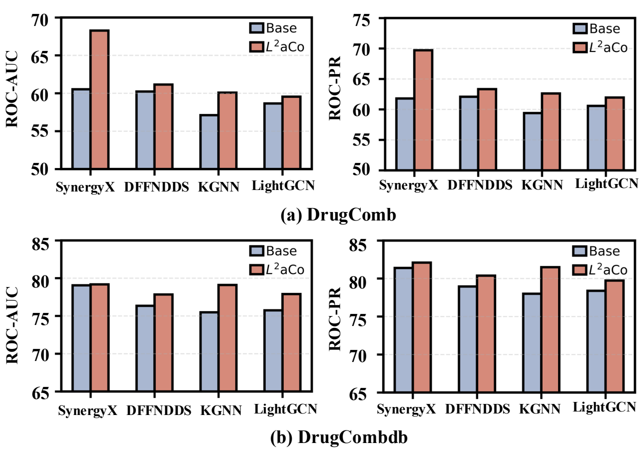


**Figure 4: Performance under long-tail cell-line setting. (a) DrugComb, (b) DrugCombDB.**

structured information is incorporated, L$^2$aCo+KGNN consistently outperforms standalone LLM embeddings. On DrugComb, AUC improves from 61.24 to 65.05 and AUPR from 63.07 to 66.78, with similar gains observed on DrugCombDB. The full model, L$^2$aCo+SynergyX, achieves the best performance across all settings. Compared with the strongest LLM baseline (GPT-5), it improves AUC/AUPR by 5.33/4.57 on DrugComb and by 4.33/4.46 on DrugCombDB. These consistent gains demonstrate that L$^2$aCo effectively amplifies the utility of LLM semantics through structured interaction modeling, rather than relying on embeddings alone.

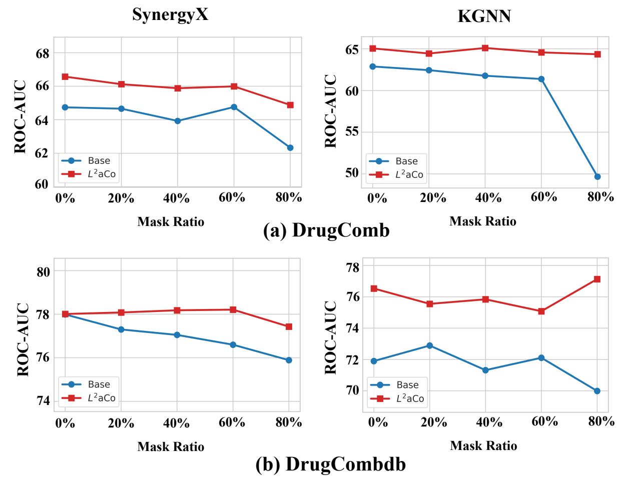


**Figure 5: Performance under perturbed semantics. ROC-AUC vs. corruption ratio (0–80%) on DrugComb (top) and DrugCombDB (bottom). SynergyX is shown on the left and KGNN on the right.**

#### 4.4.3 *Performance on Perturbed Semantics (Q4).*

We further evaluate L$^2$aCo with controlled input corruptions: we randomly mask drug/cell embeddings for feature-based backbones and shuffle graph edges for graph-based backbones. This setting quantifies performance degradation under missing or noisy signals and examines whether L$^2$aCo mitigates it via LLM-derived profiling and cross-view alignment.

Feature-level Representation Masking. We independently mask the input embeddings of each drug and the cell line at ratios of 20%, 40%, 60%, and 80%, simulating missing or incomplete modalities in data-scarce settings. We report ROC-AUC at each corruption level and compare each backbone with and without L$^2$aCo. As the masking ratio increases, performance degrades as expected, while L$^2$aCo consistently reduces the drop, indicating improved robustness under partial signal loss. This suggests that LLM-derived drug/cell profiles contribute complementary semantics that compensate for missing features, and that cross-view alignment injects relational knowledge to stabilize representations.

Topology-level Edge Shuffling for GNN Backbones. For GNN-based backbones, we add structural noise by randomly shuffling edges while keeping the total edge count fixed, and compare models with and without L$^2$aCo using ROC-AUC. Although edge shuffling degrades performance overall, L$^2$aCo consistently mitigates the decline, suggesting that LLM-derived profiles and cross-view alignment provide robust auxiliary signals when structural evidence is corrupted. As shown in Figure 5, on DrugComb the base model shows a sharp ROC-AUC drop under strong perturbations, whereas L$^2$aCo degrades more gradually, indicating greater stability and generalization under severe noise or information loss. As perturbations intensify, performance decreases as expected; however, the drop with L$^2$aCo is smaller and smoother, suggesting it compensates for missing semantics through LLM-derived profiles and cross-view alignment.

Table 2: Performance of L$^2$aCo under different large language models (i.e., GPT-3.5-turbo, GPT-4o-mini, GPT-5)

| Model | Variants | ACC | ROC-AUC | ROC-PR |
|---|---|---|---|---|
| KGNN | base | 65.08±0.57 | 71.90±0.47 | 76.70±0.50 |
| | w/ GPT-3.5 | 68.16±0.26 | 74.45±0.34 | 78.50±0.50 |
| | w/ GPT-4o-mini | 70.38±0.49 | 76.16±0.41 | 79.60±0.48 |
| | w/ GPT-5 | 69.33±0.82 | 76.52±0.85 | 80.76±0.94 |
| | **best impr. (%)** | **5.30%** | **4.62%** | **4.06%** |
| LightGCN | base | 65.66±0.09 | 71.28±0.07 | 76.18±0.10 |
| | w/ GPT-3.5 | 69.29±0.58 | 76.57±0.56 | 80.67±0.90 |
| | w/ GPT-4o-mini | 68.20±0.48 | 74.24±0.75 | 78.41±0.34 |
| | w/ GPT-5 | 70.30±0.82 | 76.72±1.06 | 80.53±1.07 |
| | **best impr. (%)** | **7.07%** | **7.63%** | **5.89%** |
| H-GAT | base | 66.04±0.41 | 72.31±0.49 | 77.27±0.27 |
| | w/ GPT-3.5 | 67.50±0.52 | 73.70±1.06 | 78.07±0.83 |
| | w/ GPT-4o-mini | 67.05±0.35 | 73.50±0.89 | 77.79±0.03 |
| | w/ GPT-5 | 67.84±0.47 | 73.98±0.77 | 78.18±0.51 |
| | **best impr. (%)** | **2.73%** | **2.31%** | **1.18%** |

Table 3: Performance of different LLMs embeddings for drug combination recommendation.

| LLM | DrugComb | | DrugCombDB | |
|---|---|---|---|---|
| | ROC-AUC | ROC-PR | ROC-AUC | ROC-PR |
| GPT-5 | 61.24±0.09 | 63.07±0.23 | 73.68±0.82 | 78.68±0.12 |
| GPT-3.5-turbo | 63.35±0.31 | 65.16±0.46 | 73.52±0.57 | 78.41±0.55 |
| GPT-4o-mini | 61.83±0.11 | 64.01±0.10 | 70.64±0.12 | 76.51+0.09 |
| L$^2$aCo+KGNN | 65.05±0.41 | 66.78±0.43 | 76.53±0.85 | 80.76±0.94 |
| L$^2$aCo+SynergyX | 66.57±0.57 | 67.64±0.70 | 78.01±0.42 | 83.14±0.67 |

## 4.5 Case Study

To evaluate whether L$^2$aCo enables learning beyond the scope of experimental data, we present a case study on the long-tail cell line PC-3 using a drug pair (Amifostine and Cabazitaxel) known to exhibit synergistic effects. Despite the expected high synergy, limited experimental coverage forces the base model to rely on sparse drug–cell interaction signals, resulting in a low predicted synergy score. This discrepancy highlights a key limitation of purely data-driven methods: learned representations are confined to observed experimental patterns, limiting reasoning over mechanistically related interactions. As shown in Figure 6, L$^2$aCo enhances the cell representation with LLM-derived semantic knowledge grounded in biomedical context. The generated description captures key molecular features of PC-3, such as PTEN loss and PI3K/AKT pathway activation, and links these traits to mechanistic compatibility between taxane-like ester-rich structures (Cabazitaxel) and thiol or phosphorothioate functional groups (Amifostine). Through cross-view alignment, L$^2$aCo integrates these semantically inferred relationships into the learned representation space, enabling the model to incorporate knowledge not directly available from experimental data. Consequently, L$^2$aCo assigns a synergy score that better aligns with the known effectiveness of this drug combination while providing an interpretable semantic rationale. This case demonstrates that L$^2$aCo learns richer representations that support reasoning

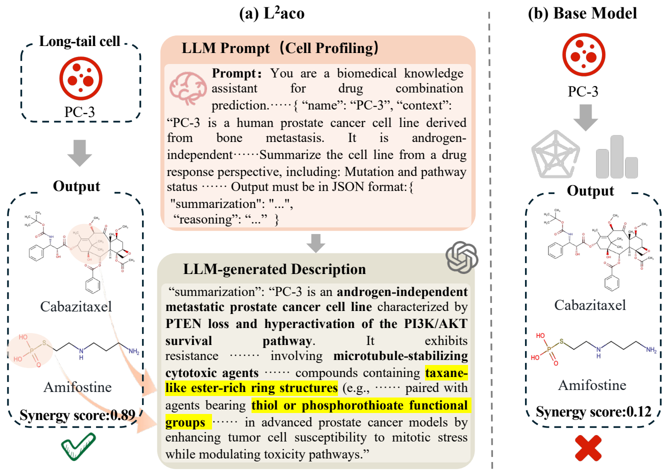


Figure 6: L$^2$aCo leverages LLM-derived semantics to improve synergy prediction for the long-tail cell line PC-3.

beyond observed experimental data, particularly for long-tail cell lines with limited experimental evidence.

## 5 Conclusion

This paper presents L$^2$aCo, a model-agnostic framework that leverages large language models to enhance representations for drug combination recommendation. By enriching drug and cell-line embeddings with LLM-inferred biomedical knowledge and aligning unstructured semantics with structured signals, L$^2$aCo improves generalization beyond limited experimental data. Experiments show consistent gains over strong baselines, especially on novel and long-tail cell lines, while remaining robust to noisy knowledge and variations in the profiling LLM. Future work will explore more reliable profiling, uncertainty-aware alignment, and extensions to multi-omics and clinically relevant generalization.

## 6 Limitations and Ethical Considerations

This study uses only publicly available, de-identified data for research purposes. L$^2$aCo is evaluated on preclinical and public datasets without real-world clinical data, which may limit its applicability to complex clinical settings and patient variability.

## GenAI Disclosure

Generative AI tools were used solely for language polishing and improvement of writing clarity and organization.

## Acknowledgments

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# 7 Appendix

## A Prompt Template and Additional Examples

### A.1 Prompt Template

L$^2$aCo constructs structured prompts to elicit semantic knowledge from large language models (LLMs) in a consistent and task-relevant manner. Rather than relying on free-form queries, we adopt a lightweight and modular prompt template that separates *entity description* from *knowledge elicitation objectives*.

*Drug Prompt Template.* Given a drug entity $d$, the prompt template is defined as:

```
You will serve as an assistant for predicting drug
     combinations. You need help me summarize the
     drugs, including its mechanism of action,
     target pathways, metabolic pathway, common side
      effects, and the types of diseases it is
     typically used to treat.
Here are the instructions:
1. I will provide you with information in the form
     of a JSON string that describes the drug:
{
    "name": "the name of the drug",
    "context": "the related facts extracted from
     knowledge base"
}

Requirements:
1. Please provide your answer in JSON format,
     following this structure:
{
    "summarization": "A summarization of the drug"
    "reasoning": "briefly explain your reasoning for
      the summarization"
}
2. Please ensure that the "summarization" is no
     longer than 200 words.
3. Please ensure that the "reasoning" is no longer
     than 200 words.
4. Do not provide any other text outside the JSON
     string.
```

*Cell Line Prompt Template.* For a cell line entity $c$, the prompt template is defined as:

```
You will serve as an assistant for predicting drug
     combinations. You need help me summarize the
     cell lines, covering its gene expression
     profile, mutation status, drug sensitivity
     profile, and its applications in drug research.
Here are the instructions:
1. I will provide you with information in the form
     of a JSON string that describes the drug:
{
    "name": "the name of the cell line",
    "context": "the related facts extracted from
     knowledge base"
}

Requirements:
1. Please provide your answer in JSON format,
     following this structure:
{
    "summarization": "A summarization of the drug"
    "reasoning": "briefly explain your reasoning for
      the summarization"
}
2. Please ensure that the "summarization" is no
     longer than 200 words.
3. Please ensure that the "reasoning" is no longer
     than 200 words.
4. Do not provide any other text outside the JSON
     string.
```

These templates are intentionally generic and entity-centric, allowing the LLM to infer biologically meaningful semantics without introducing task-specific labels or supervision signals.

—

### A.2 Additional Examples

*Drug Example.* For the drug *Trametinib*, the constructed prompt encourages the LLM to identify its role as a MEK inhibitor and summarize its involvement in the MAPK signaling pathway. The generated semantic summary highlights its regulatory effects on cell growth and proliferation, while the rationale emphasizes its relevance in targeted therapies. This information is not explicitly encoded in experimental feature vectors but is captured through LLM-derived semantic representations.

*Cell Line Example.* For a melanoma cell line characterized by aberrant MAPK pathway activation, the cell prompt guides the LLM to summarize its transcriptional profile and biological state. The resulting semantic representation encodes functional context, such as proliferative signaling and pathway dependencies, which complements sparse or noisy gene expression measurements.

*Relation-Level Example.* In addition to entity-level semantics, L$^2$aCo implicitly captures relational knowledge. For example, when provided with prompts describing a drug and a disease context, the LLM-generated summaries often reflect known associations between molecular targets and disease pathways. Such relational semantics can be viewed as soft, implicit edges that complement structured biomedical knowledge graphs.

—

## A.3 Discussion

The above examples illustrate that LLM-derived semantic representations encode high-level biomedical knowledge that is difficult to obtain from experimental data or curated knowledge graphs alone. By aligning these semantic representations with experimental features, $L^2$aCo effectively transfers external knowledge into downstream prediction models. This mechanism explains the observed robustness of $L^2$aCo under semantic masking and knowledge graph perturbation experiments.

## B. Additional Experiments

## Implementation details

*Details of $L^2$aCo.* We use GPT-5 to generate structured biomedical profiles for drugs and cell lines using a fixed prompt template and schema-constrained decoding (temperature = 0.1, max tokens = 1024). The generated texts are encoded with *text-embedding-3-large* to obtain semantic embeddings.

$L^2$aCo aligns semantic and experimental representations using an InfoNCE-style contrastive objective with cosine similarity and in-batch negatives. We apply $L_2$ normalization and use an alignment temperature $\tau = 0.07$.

We evaluate under three split settings: *random*, *novel cell-line*, and *long-tail cell-line*, corresponding to standard, unseen-cell-line, and tail-distribution generalization settings, respectively. Detailed split statistics are provided in the repository.

*Details of baselines.* We use the official implementations for all baselines, including **SynergyX**[3], **DeepDDS**[4], **DFFNDDS**[5], **HanSynergy**[6], **KGNN**[7], **LightGCN**[8], and **H-GAT**[9]. We follow the default configurations provided in the original repositories for all baselines.

## More results

*Performance on common cell lines.* Under the common cell-line setting, we evaluate $L^2$aCo under the random-split protocol. Tables 4 show consistent improvements on DrugComb across three representative backbones (SynergyX, DFFNDDS, and DeepDDS) and all evaluation metrics. This suggests that the gains are not an artifact of a specific architecture, but stem from a more general effect: LLM-derived semantics provide complementary, high-level signals that are weakly correlated with the experimental feature space and therefore improve discrimination beyond what purely data-driven representations capture. The fact that improvements persist across datasets further indicates that semantic augmentation is less sensitive to dataset-specific biases and helps stabilize learning under different splits. Overall, these results support $L^2$aCo as a backbone-agnostic and transferable plug-in for drug synergy prediction.

[3]https://github.com/GSanShui/SynergyX
[4]https://github.com/Sinwang404/DeepDDs
[5]https://github.com/sorachel/DFFNDDS
[6]https://github.com/cn0403/HANSynergy
[7]https://github.com/xzenglab/KGNN
[8]https://github.com/kuandeng/LightGCN
[9]https://github.com/dmlc/dgl/tree/master/examples/pytorch/rgat

**Table 4: Performance on DrugComb under the random split setting. We report ACC, ROC-AUC and ROC-PR (mean ± std) for different models. impr. (%) denotes the relative improvement of $L^2$aCo over the base setting.**

| Model | Setting | DrugComb | | |
|---|---|---|---|---|
| | | ACC | ROC-AUC | ROC-PR |
| SynergyX | base | 73.80 ± 0.07 | 81.34 ± 0.12 | 82.81 ± 0.38 |
| | $L^2$aCo | 74.43 ± 0.16 | 82.12 ± 0.26 | 83.58 ± 0.59 |
| | **impr. (%)** | +0.85% | +0.96% | +0.93% |
| DFFNDDS | base | 71.15 ± 0.20 | 79.08 ± 0.07 | 81.64 ± 0.14 |
| | $L^2$aCo | 71.59 ± 0.11 | 79.64 ± 0.17 | 81.99 ± 0.17 |
| | **impr. (%)** | +0.62% | +0.71% | +0.43% |
| DeepDDS | base | 71.24 ± 0.38 | 79.74 ± 0.26 | 84.66 ± 0.59 |
| | $L^2$aCo | 73.10 ± 0.52 | 81.10 ± 0.74 | 85.34 ± 0.46 |
| | **impr. (%)** | +2.61% | +1.71% | +0.80% |

*Permance of $L^2$aCo with KL-based alignment.* We further assess KL-based alignment for injecting LLM-derived semantic representations into downstream synergy predictors. As shown in Table 5, KL alignment delivers consistent gains across most backbones on both DrugComb and DrugCombDB, with larger improvements on DrugCombDB. This is likely because KL acts as a distribution-level calibrator: since semantic and experimental features originate from different sources, naive fusion can introduce scale and confidence shifts and amplify noise; KL alignment encourages cross-modal consistency, stabilizing the semantic signal and improving cross-dataset generalization. It benefits strong backbones via regularization and helps graph models avoid propagating semantic noise, which is often reflected more clearly in ROC-AUC/ROC-PR. Compared with CL-based enhancement, which mainly increases semantic discriminability and can be sensitive to sampling design, KL emphasizes cross-modal consistency and is typically more stable under dataset shifts.

*Accuracy performance for long-tail cell lines.* Under the long-tail cell-line setting, we deliberately focus on cell lines with scarce interactions, where models are more susceptible to sparse supervision and noisy signals, better reflecting the long-tail distribution encountered in practice. As shown in Figure 7, on both DrugComb and DrugCombDB, $L^2$aCo consistently achieves higher ACC than the base setting across multiple backbones. These results suggest that, in the tail regime, structured experimental features alone are often insufficient to capture the biological context of rare cell lines, whereas LLM-derived semantic representations provide higher-level, transferable priors that mitigate representation degradation under limited data and improve predictive accuracy for under-represented cell lines.

*Robustness of $L^2$aCo.* We assess the robustness of $L^2$aCo under controlled feature corruption by randomly masking drug and cell-line embeddings, which simulates missing or incomplete modalities

**Table 5: Performance comparison between original models and models augmented with KL-based augmentation on the DrugComb and DrugCombDB datasets. Results of augmented models are reported as mean ± standard deviation over five random seeds.**

| Model | Setting | DrugComb | | | DrugCombDB | | |
|---|---|---|---|---|---|---|---|
| | | ACC | ROC-AUC | ROC-PR | ACC | ROC-AUC | ROC-PR |
| SynergyX | base | 60.66 ± 0.73 | 64.74 ± 0.62 | 66.10± 0.59 | 71.26± 0.67 | 77.99± 0.48 | 82.90± 0.73 |
| | $L^2$aCo | 61.47± 0.27 | 65.52 ± 0.36 | 67.21± 0.39 | 71.97± 0.34 | 78.45 ± 0.53 | 84.55± 0.41 |
| | **impr. (%)** | +1.34% | +1.20% | +1.68% | +1.00% | +0.59% | +1.99% |
| DFFNDDS | base | 58.93 ± 0.18 | 61.50 ± 0.09 | 63.13 ± 0.14 | 71.06 ± 0.23 | 76.56 ± 0.15 | 78.91 ± 0.54 |
| | $L^2$aCo | 60.93 ± 2.26 | 64.21 ± 2.77 | 64.42 ± 2.95 | 72.73 ± 2.54 | 79.53 ± 3.54 | 83.15 ± 3.29 |
| | **impr. (%)** | +0.39% | +0.44% | +0.24% | +0.23% | +0.38% | +0.53% |
| DeepDDS | base | 55.74 ± 2.94 | 61.54 ± 1.28 | 64.46 ± 0.73 | 64.02 ± 4.12 | 85.51 ± 0.67 | 99.72 ± 0.04 |
| | $L^2$aCo | 55.85 ± 0.71 | 61.43 ± 0.76 | 66.17 ± 0.74 | 68.78 ± 5.66 | 89.85 ± 1.74 | 99.80 ± 0.03 |
| | **impr. (%)** | +0.20% | -0.18% | +2.65% | +7.44% | +5.08% | +0.08% |
| LightGCN | base | 59.88 ± 0.89 | 62.88 ± 0.08 | 63.23 ± 0.12 | 65.67 ± 0.09 | 71.28 ± 0.07 | 76.18 ± 0.10 |
| | $L^2$aCo | 59.79 ± 0.43 | 64.20 ± 0.43 | 65.66 ± 0.80 | 69.94 ± 0.43 | 76.85 ± 0.26 | 80.62 ± 0.41 |
| | **impr. (%)** | -0.15% | +2.10% | +3.84% | +6.50% | +7.81% | +5.83% |
| H-GAT | base | 58.22 ± 0.53 | 62.56 ± 0.15 | 62.73 ± 0.21 | 66.04 ± 0.41 | 72.31 ± 0.49 | 77.27 ± 0.27 |
| | $L^2$aCo | 59.00 ± 0.26 | 64.09 ± 0.73 | 65.63 ± 1.21 | 65.87 ± 0.26 | 72.13 ± 0.17 | 76.66 ± 0.17 |
| | **impr. (%)** | +1.34% | +2.45% | +4.62% | -0.26% | -0.25% | -0.79% |

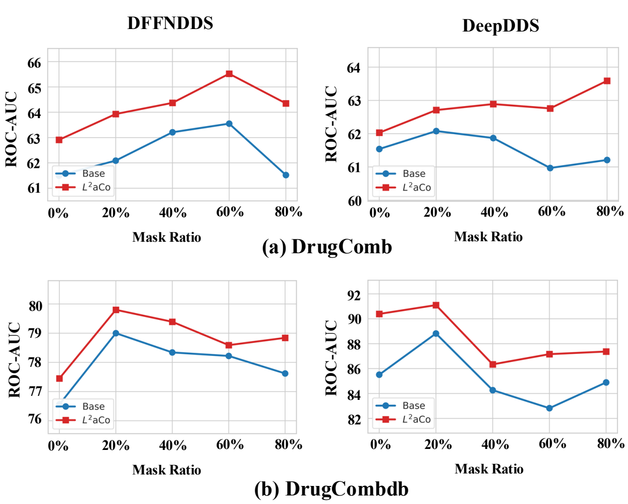


**Figure 8: : Performance under perturbed semantics. ROC-AUC vs. corruption ratio (0–80%) on DrugComb (top) and Drug- CombDB (bottom). DFFNDDS is shown on the left and DeepDDS on the right.**

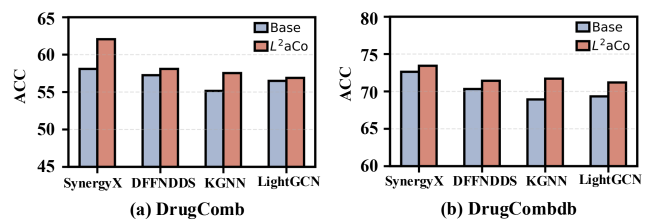


**Figure 7: ACC performance under the long-tail cell-line setting.**

in practice. In the long-tail cell line setting, we report ROC-AUC for DFFNDDS and DeepDDS under mask ratios from 0% to 80% on both DrugComb and DrugCombDB. As shown in Figure 8, increasing masking leads to expected performance degradation for the base models, while $L^2$aCo consistently mitigates the drop and maintains a higher ROC-AUC across corruption levels. This indicates that semantic profiling provides complementary, transferable signals that are particularly valuable when experimental supervision is sparse and noisy. We additionally evaluate the common cell line setting with mask ratios from 0% to 60% on SynergyX, DFFNDDS, and DeepDDS (ACC with error bars). Figure **??** shows that $L^2$aCo remains consistently more stable than the corresponding backbones under increasing corruption, exhibiting smaller performance variance and a smoother degradation trend. Overall, these results suggest that $L^2$aCo improves robustness to partial signal loss by stabilizing representation learning: LLM-derived profiles compensate for missing features, while cross-view alignment regularizes the fused representation and reduces sensitivity to feature corruption.